Memory Production under Coarse-Graining: A Balance Law for the Indivisibility Deficit*

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July 2026

Abstract

Eliminating variables from a Markov process generically produces memory: the reduced transition family violates the Chapman–Kolmogorov property although the full family satisfies it. This paper quantifies that production using the liftability profile introduced for transition families without a generator: within a declared dilation category, $h_d(V)$ is the attained distance from V to the compressions of d-state flat transports, and the charge of a family observed on n states is the dimension-matched value $h_n(V)$, the metric size of its Chapman–Kolmogorov failure. A natural conjecture is corrected first: the charge is not monotone under coarse-graining of the state space. We give elementary counterexamples in both directions: a lumped Markov chain whose reduced family carries positive charge although the fine family carries none, and a product family whose memory-bearing factor is discarded by the reduction, so that coarse-graining destroys the charge. What holds instead is a balance law, and it is exact. For a lumping compatible with the law of the process, the fine deficit splits orthogonally into a flux term, its compression through the cut, which is contractive in the induced metric, and a severed remainder that no coarse observable retains; the coarse deficit is the flux plus a production term sourced at the cut itself. Production vanishes exactly when the cut is dynamically closed on the directions the process reaches, recovering weak lumpability in its two-time formulation as the zero set for Markov fine dynamics, and the charge inherits the balance law. The production term is the fluctuation current across the cut, a form of the Mori–Zwanzig memory kernel independent of the choice of projection operator, and, when the fine family is flat, its size is measured two-sidedly by the residual at the metric projection onto flat families. Monotonicity survives in exactly the form the balance law permits. Along a tower of state-space coarse-grainings in which each cut severs nothing and each production term is coherent with the transmitted deficit, so that in particular no memory-bearing autonomous factor is discarded, the deficit norm cannot decrease; both hypotheses are sharp, since discarding a memory-bearing autonomous factor severs its charge in full, while an explicit perfect-interference construction that severs nothing annihilates the charge by setting production exactly opposite to the transmitted deficit. Finite-state examples are computed explicitly, in rational arithmetic throughout.

Keywords. Chapman–Kolmogorov divisibility, lumpability, coarse-graining, memory kernel, Mori–Zwanzig formalism, indivisible stochastic processes, balance law

AMS subject classifications. 60J10, 82C31, 60G05, 15B51

*This work was supported by Adagio Institute, Inc.

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Significance statement

Every reduced description of a dynamical system is obtained by a cut: some variables are kept, the rest are eliminated. Statistical mechanics has known since Nakajima, Zwanzig, and Mori that the eliminated variables return as memory, and Markov chain theory has known since Kemeny and Snell that a lumped chain generically fails the Chapman–Kolmogorov property. This paper joins the two facts into bookkeeping. The size of the Chapman–Kolmogorov failure, the charge of the companion projection theory, is shown to obey an exact balance law across any cut: the fine deficit splits orthogonally into a transmitted part and a severed part, and the coarse deficit is the transmitted part plus a production term sourced at the cut. Memory is therefore not a substance that only accumulates under reduction. It is produced at cuts that are not dynamically closed, destroyed by cuts that sever the directions carrying it, and conserved exactly when neither happens. The production term is a Mori–Zwanzig memory kernel written without choosing a projection operator, and, for Markov fine dynamics, the zero set of production recovers weak lumpability in its two-time formulation. The balance law converts the folklore that "coarse-graining creates memory" into a theorem with an exact exception set, and it equips the indivisibility program with the tool it needs next: a way to trace where the memory measured in data was made.

1 Introduction

Let $X = (X_{t_i})_{i=0}^{m}$ be a stochastic process on a finite state space $[N]$, observed on a finite time grid, and let V denote its two-time transition family, $V(j \leftarrow i)_{vu} = \mathbb{P}(X_{t_j} = v \mid X_{t_i} = u)$. If X is Markov the family composes,

$$V(t \leftarrow s) \;=\; V(t \leftarrow r)\, V(r \leftarrow s), \qquad s \le r \le t, \tag{1}$$

the Chapman–Kolmogorov property. Now eliminate variables: fix a surjection $\lambda : [N] \to [n]$ with $n < N$ and observe only $Y = \lambda(X)$. The transition family of Y generically violates (1) even when the family of X satisfies it [5, 2]. The violation is the deficit

$$\Delta_V(t, s, r) \;=\; V(t \leftarrow s) - V(t \leftarrow r)\, V(r \leftarrow s), \qquad s < r < t, \tag{2}$$

the central object of the program of which this paper is a part [14, 15, 16, 17, 18], and, in statistical mechanics, the shadow at the level of two-time conditionals of the memory kernel that projection-operator techniques attach to any reduced description [9, 21, 8, 4, 22, 19]. The companion paper [18] converted the deficit into an invariant: within a declared dilation category the liftability profile $h_d(V)$ is the attained distance from V to the compact model set $\mathcal{F}_d$ of compressions of d-state flat transports, and at the dimension-matched value $d = n$ the profile is the distance to the flat families themselves,

$$h_n(V) \;=\; \operatorname{dist}(V, \mathcal{F}\mathrm{lat}_n), \tag{3}$$

the *charge* of the family: the metric size of its Chapman–Kolmogorov failure [18, Example 5.2].[1]

This paper asks the question that the invariant makes well-posed: *how does the charge transform under coarse-graining of the state space?* Two intuitions compete. Elimination of variables creates memory, so the charge should grow as the description coarsens; this is the projection-operator intuition. Coarse-graining destroys information, and the charge is information about the failure of (1), so the charge should shrink; this is the data-processing intuition. Both are wrong as

[1] Numbered references to [18] follow the SSRN preprint version.

monotonicity statements, and Section 3 proves it with elementary exact counterexamples: a three-state Markov chain whose two-block lumping carries positive charge, with coarse deficit of metric size exactly $\sqrt{18/23575}$, although the fine family carries none, and a four-state product family of positive charge whose coarse-graining onto its memoryless factor is exactly flat. Coarse-graining can create charge from nothing and annihilate it entirely.

What replaces monotonicity is a balance law, and it is exact, not asymptotic. Section 4 proves that for any lumping compatible with the law of the process, the coarse deficit decomposes as

$$\Delta_{V^\lambda}(t,s,r) \;=\; \underbrace{\mathcal{C}_s\big[\Delta_V(t,s,r)\big]}_{\text{flux}} \;+\; \underbrace{\mathrm{P}_\lambda(t,s,r)}_{\text{production}}, \tag{4}$$

where the flux term is the compression of the fine deficit through the cut and the production term is sourced at the cut itself. The two terms have opposite characters. The compression $\mathcal{C}$ is a conditional expectation in the metric weighted by the law of the process (Lemma 2.3); it is therefore contractive in the induced metric, and its contraction defect is itself an invariant: the fine deficit splits orthogonally into a transmitted component and a *severed* component that no coarse observable retains,

$$\|\Delta_V\|^2 \;=\; \|\mathcal{C}[\Delta_V]\|^2 + \|\Xi_\lambda\|^2. \tag{5}$$

The production term is non-negative in the only sense a matrix-valued source can be, through its metric size, and its vanishing is completely characterized: production is zero exactly when the cut is dynamically closed, meaning the conditional law of the coarse future given the fine present depends on the present only through its coarse part, on the directions the process actually reaches (Theorem 4.5). For a fine Markov family the zero set of production recovers weak lumpability in its two-time formulation, with strong lumpability as a strict sufficient condition [5, 12, 13].

Section 5 identifies what the production term is. Writing Π_r for the conditional-expectation projection onto coarse observables at time t_r and $Q_r = I - \Pi_r$ for its fluctuation complement, the production is

$$\mathrm{P}_\lambda(t,s,r) \;=\; L\,V(t \leftarrow r)\,Q_r\,V(r \leftarrow s)\,R_s :$$

propagate from s to the cut time r, keep only the component invisible to the coarse description, propagate on to t, observe. This is the two-time Mori–Zwanzig memory kernel, and the discrete Dyson expansion of Proposition 5.1 exhibits the full generalized Langevin structure of the coarse family, with the flat part as the all-resolved word and the memory kernels as the fluctuation runs [3, 6]. The formalism ordinarily depends on a choice of projection operator; here it does not, and Theorem 5.2 states the precise sense: the deficit of the intrinsic coarse family is a functional of observed quantities alone, the memory kernel built from the law projection reproduces it exactly, and every other admissible projection produces a kernel differing from the deficit by the mismatch of its resolved model with the observed conditionals. The residual at the metric projection of V^λ onto $\mathcal{F}\mathrm{lat}_n$ is then the canonical, projection-free representative of the memory kernel, and the charge measures it two-sidedly through the Łojasiewicz comparison of [18, Proposition 5.3].

Section 6 treats towers of coarse-grainings and proves the monotonicity theorem the counterexamples leave room for: along a tower step that severs nothing and whose production is coherent with the transmitted deficit, the deficit norm cannot decrease, and it strictly increases whenever production is present (Theorem 6.2). Both hypotheses are sharp, and both sharpness witnesses are exact. Discarding an autonomous factor is the general mechanism of total severing: Proposition 6.4 shows that projecting away an independent factor is always a dynamically closed cut whose flux is exactly the retained factor's deficit, so the discarded factor's memory is severed in full, which is the destruction counterexample run in general form. And coherence cannot be dropped even

when nothing at all is severed: Proposition 6.5 constructs a three-state process, with its joint law displayed and strictly positive, whose two-block cut severs nothing, discards no factor, and yet has production exactly opposite to the transmitted deficit, so that the coarse family is flat and the charge is annihilated by pure destructive interference. Section 7 gathers the four worked families of the paper, one for each pure regime of the balance law, with every entry an exact rational number. Section 8 states open problems.

The contributions, in the order of the paper: (i) exact counterexamples to monotonicity of the charge in both directions (Theorem 3.1); (ii) the compression-as-conditional-expectation lemma and the induced-metric geometry of cuts (Lemma 2.3); (iii) the exact balance law with the conservation identity for the severed component (Theorem 4.2); (iv) the characterization of zero production by dynamical closure and the recovery of weak lumpability (Theorem 4.5, Corollary 4.6); (v) the identification of production as a projection-free Mori–Zwanzig memory kernel with the discrete Dyson expansion (Section 5); (vi) tower monotonicity with sharp hypotheses, the autonomy proposition, and the exact perfect-interference construction (Section 6); and (vii) explicit finite-state computations, in rational arithmetic throughout, realizing each pure regime of the balance law (Section 7). The paper contains no numerical optimization, no simulation, and no data: every displayed quantity is an exact rational number or the square root of one, verifiable by hand.

2 Setting: families, the law metric, and cuts

2.1 Families and the charge

Notation follows [18]. Fix a finite time grid $\mathbb{T} = \{t_0 < \cdots < t_m\}$, $m \geq 2$, written $0, \ldots, m$. For an integer $k \geq 2$ let $\mathcal{S}_k$ be the column-stochastic $k \times k$ matrices, and let $\mathcal{V}_k = \prod_{i<j} \mathcal{S}_k$ be the transition families on k states, with elements $V = \{V(j \leftarrow i)\}_{i<j}$. A family is *flat* if it satisfies (1) on the grid; the set of flat families is $\mathcal{F}\mathrm{lat}_k \subset \mathcal{V}_k$, the zero set of the deficit (2). A family is *process-induced* if it is the two-time conditional family of a process on $[k]$ with strictly positive one-time laws.

Throughout, a family carries its *laws*: strictly positive probability vectors $p_0, \ldots, p_m$ on $[k]$, supplied by the process when the family is process-induced ($V(j \leftarrow i)\, p_i = p_j$ then holds for all $i < j$). The laws enter the metric. For kernel differences A, B attached to the pair $(i < j)$ define

$$\langle A, B\rangle_{(j\leftarrow i)} \;=\; \sum_{v,u=1}^{k} \frac{p_i(u)}{p_j(v)}\, A_{vu} B_{vu}, \qquad \|A\|_{(j\leftarrow i)} = \langle A, A\rangle_{(j\leftarrow i)}^{1/2}, \tag{6}$$

and for families the sum over pairs with unit pair weights; deficits $\Delta(t, s, r)$ carry the $(t \leftarrow s)$ weights, and deficit cochains the sum over triples. The weights (6) lie in the admissible class of [18]: they are strictly positive entry weights determined by the law of the process, so by [18, Lemma 4.8] the zero sets of all distances below, and hence every qualitative statement of this paper, are independent of the choice within the class, while (6) is the choice under which compression becomes a conditional expectation (Lemma 2.3), which is why we fix it. The *charge* of a family is its dimension-matched liftability value (3), $h_n(V) = \mathrm{dist}(V, \mathcal{F}\mathrm{lat}_n)$ in the metric (6); by [18, Proposition 5.3] there are constants $L_*, C_* > 0$ and $\theta \in (0, 1]$, depending on the dimension, the grid, and the metric weights fixed by the laws, with

$$\tfrac{1}{L_*}\, \|\Delta_V\| \;\leq\; h_n(V) \;\leq\; C_*\, \|\Delta_V\|^{\theta}, \tag{7}$$

so the charge and the deficit norm vanish together and each controls the other on the compact $\mathcal{V}_n$. The balance law of this paper is exact at the level of the deficit cochain; it transfers to the charge through (7), and we will say so explicitly at each transfer.

2.2 Cuts: lumpings, compression, disintegration

Definition 2.1 (Cut)**.** A *cut* of $[N]$ is a surjection $\lambda : [N] \to [n]$, $2 \le n < N$, encoded by the 0/1 matrix $L \in \mathbb{R}^{n\times N}$, $L_{au} = \mathbf{1}\{\lambda(u) = a\}$; note $\mathbf{1}^\top L = \mathbf{1}^\top$. The cut is *compatible with the law* of a family (V, p) if all p_i are strictly positive, which we assume throughout; then the block masses $p_i^\lambda = Lp_i$ are strictly positive. The *disintegration* of p_i over the blocks is the column-stochastic section $R_i \in \mathbb{R}^{N\times n}$,

$$(R_i)_{ua} \;=\; \frac{p_i(u)\,\mathbf{1}\{\lambda(u) = a\}}{p_i^\lambda(a)}, \qquad LR_i = I_n,$$

and $\Pi_i = R_iL \in \mathbb{R}^{N\times N}$ is the induced idempotent, $\Pi_i^2 = \Pi_i$, with fluctuation complement $Q_i = I - \Pi_i$. The *compression* of the family is

$$V^\lambda(j \leftarrow i) \;=\; L\,V(j \leftarrow i)\,R_i,$$

with laws p_i^λ; the compression of a kernel difference A at the pair $(i < j)$ is $\mathcal{C}_i[A] = LAR_i$.

Proposition 2.2 (The compression is the observed family)**.** *If V is the conditional family of a process X with laws p, then V^λ is the conditional family of the lumped process $Y = \lambda(X)$:*

$$V^\lambda(j \leftarrow i)_{ba} \;=\; \mathbb{P}\big(\lambda(X_{t_j}) = b \mid \lambda(X_{t_i}) = a\big),$$

with laws p_i^λ. In particular V^λ is process-induced and marginal-consistent, and its charge is the charge of the process actually observed through the cut.

Proof. $\big(LV(j \leftarrow i)R_i\big)_{ba} = \sum_{v:\lambda v=b}\ \sum_u\ V(j \leftarrow i)_{vu}\,\frac{p_i(u)\mathbf{1}\{\lambda u=a\}}{p_i^\lambda(a)} = \frac{\mathbb{P}(\lambda X_{t_j}=b,\ \lambda X_{t_i}=a)}{\mathbb{P}(\lambda X_{t_i}=a)}$ by the law of total probability. Marginal consistency $V^\lambda(j \leftarrow i)p_i^\lambda = p_j^\lambda$ follows from $V(j \leftarrow i)p_i = p_j$ and $R_ip_i^\lambda = \Pi_ip_i = p_i$. □

This is the classical setting in which the Chapman–Kolmogorov property fails [5, 2], and the compression coincides with the observable compression κ of the dilation theory of [18]: coarse families are exactly the objects whose distance to $\mathcal{F}\mathrm{lat}_n$ the charge measures.

2.3 Compression as conditional expectation

The geometric engine of the paper is the following elementary lemma; it is the reason the metric (6) is the right one for cuts.

Lemma 2.3 (Conditional expectation, isometric lift, Pythagoras)**.** *Fix a pair $(i < j)$ and identify a kernel difference A at $(i < j)$ with its density $f_A(v, u) = A_{vu}/p_j(v)$ in $L^2(p_j \otimes p_i)$; the identification is an isometry for* (6)*. Let $\mathcal{B}_\lambda$ be the sub-σ-algebra of $[N] \times [N]$ generated by $\lambda \times \lambda$. Then:*

(a) the density of the compression $\mathcal{C}_i[A] = LAR_i$, taken with respect to the coarse laws, is the conditional expectation: $f_{\mathcal{C}_i[A]} \circ (\lambda \times \lambda) = \mathbb{E}_{p_j\otimes p_i}[f_A \,|\, \mathcal{B}_\lambda]$;

(b) the block lift $\iota_i[\delta]_{vu} = \delta_{\lambda v,\lambda u}\ p_j(v)/p_j^\lambda(\lambda v)$ is an isometric right inverse of $\mathcal{C}_i$ onto the $\mathcal{B}_\lambda$-measurable kernel differences, and it preserves vanishing column sums;

(c) consequently $\mathcal{C}_i$ is a metric surjection onto the coarse kernel differences in the induced metric*, the metric* (6) *built from the lumped laws: it is contractive, $\|\mathcal{C}_i[A]\| \le \|A\|$, and exactly Pythagorean,*

$$\|A\|^2 \;=\; \|\mathcal{C}_i[A]\|^2 \;+\; \|A - \iota_i\mathcal{C}_i[A]\|^2, \tag{8}$$

with $A - \iota_i\mathcal{C}_i[A]$ orthogonal to every block lift.

Proof. (a) For $b = \lambda v$, $a = \lambda u$,

$$\mathbb{E}[f_A|\mathcal{B}_\lambda]\,(v,u) = \frac{\sum_{v'\in\lambda^{-1}b}\sum_{u'\in\lambda^{-1}a} p_j(v')p_i(u')\,A_{v'u'}/p_j(v')}{p_j^\lambda(b)\,p_i^\lambda(a)} = \frac{\big(L\,A\,\mathrm{diag}(p_i)\,L^\top\big)_{ba}}{p_j^\lambda(b)\,p_i^\lambda(a)} = \frac{(LAR_i)_{ba}}{p_j^\lambda(b)},$$

using that $\big(LA\,\mathrm{diag}(p_i)L^\top\big)_{ba} = \sum_{v'\in\lambda^{-1}b}\sum_{u'\in\lambda^{-1}a} A_{v'u'}\,p_i(u')$ and $R_i = \mathrm{diag}(p_i)L^\top\mathrm{diag}(p_i^\lambda)^{-1}$. The right-hand side is the density of LAR_i with respect to $(p_j^\lambda, p_i^\lambda)$ pulled back along $\lambda\times\lambda$. (b) The density of $\iota_i[\delta]$ is $f_\delta\circ(\lambda\times\lambda)$, a $\mathcal{B}_\lambda$-measurable function, and $\|f_\delta\circ(\lambda\times\lambda)\|^2_{L^2(p_j\otimes p_i)} = \sum_{b,a} p_j^\lambda(b)p_i^\lambda(a)f_\delta(b,a)^2$, the coarse norm; $\mathcal{C}_i\iota_i = \mathrm{id}$ by direct computation, and $\sum_v \iota_i[\delta]_{vu} = \sum_b \delta_{b,\lambda u} = 0$ when δ has vanishing column sums. (c) Conditional expectation is the orthogonal projection of $L^2(p_j\otimes p_i)$ onto the $\mathcal{B}_\lambda$-measurable subspace [20]; (a) and (b) transport orthogonality, contraction, and (8) to kernel differences. □

Remark 2.4. Lemma 2.3(b) is the statement that coarse kernel differences, measured with the lumped laws, embed isometrically into fine kernel differences as the block-measurable subspace; "contractive in the induced metric" in the abstract refers to exactly this pair of metrics. Everything in the lemma fails for generic entry weights: the choice (6) is load-bearing, and it is available precisely because the process supplies its laws.

3 Failure of monotonicity

Theorem 3.1 (The charge is not monotone under coarse-graining)**.** *Neither inequality between $h_N(V)$ and $h_n(V^\lambda)$ holds in general: there is a process-induced family with $h_N(V) = 0$ and $h_n(V^\lambda) > 0$, and a process-induced family with $h_N(V) > 0$ and $h_n(V^\lambda) = 0$. Both occur already for $N \le 4$, $n = 2$, on the grid $\{0,1,2\}$, with rational data.*

The two constructions are Examples 3.2 and 3.3; all displayed numbers are exact.

Example 3.2 (Production: a lumped Markov chain). Take the three-state Markov chain with column-stochastic step and uniform initial law

$$M = \begin{pmatrix} 3/4 & 1/8 & 1/4 \\ 1/8 & 3/8 & 1/4 \\ 1/8 & 1/2 & 1/2 \end{pmatrix}, \qquad p_0 = \big(\tfrac{1}{3},\tfrac{1}{3},\tfrac{1}{3}\big)^\top,$$

on the grid $\{0,1,2\}$, so the fine family $V(j\leftarrow i) = M^{\,j-i}$ is flat and $h_3(V) = 0$. Cut by $\lambda(1) = \lambda(2) = a$, $\lambda(3) = b$. The chain is not strongly lumpable for this partition: the exit probabilities into $\{3\}$ from states 1 and 2 are $1/8$ and $1/2$. The coarse laws are $p_0^\lambda = (\frac{2}{3},\frac{1}{3})$, $p_1^\lambda = (\frac{5}{8},\frac{3}{8})$, $p_2^\lambda = (\frac{41}{64},\frac{23}{64})$, and the coarse family is

$$V^\lambda(1\leftarrow 0) = \begin{pmatrix} 11/16 & 1/2 \\ 5/16 & 1/2 \end{pmatrix}, \quad V^\lambda(2\leftarrow 1) = \begin{pmatrix} 29/40 & 1/2 \\ 11/40 & 1/2 \end{pmatrix}, \quad V^\lambda(2\leftarrow 0) = \begin{pmatrix} 85/128 & 19/32 \\ 43/128 & 13/32 \end{pmatrix}.$$

The single deficit is

$$\Delta_{V^\lambda}(2,0,1) = \begin{pmatrix} 3/320 & -3/160 \\ -3/320 & 3/160 \end{pmatrix}, \qquad \|\Delta_{V^\lambda}\|^2 = \frac{18}{23575}, \tag{9}$$

in the metric (6) with the coarse laws. Since $\mathcal{F}\mathrm{lat}_2$ is exactly the zero set of the deficit, $h_2(V^\lambda) > 0$; quantitatively, $h_2(V^\lambda) \ge \frac{1}{L_*}\sqrt{18/23575}$ by (7). Coarse-graining created charge from a chargeless family.

Example 3.3 (Destruction: discarding the memory-bearing factor). Let B denote the charged two-state family of Example 3.2, $B = V^\lambda$ with laws $q_i = p_i^\lambda$: it is process-induced, being the observed family of the lumped chain. Let A be the flat two-state family of the symmetric chain $M_A = \left(\begin{smallmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{smallmatrix}\right)$ at its stationary law $(\frac{1}{2}, \frac{1}{2})$. Run the two processes independently: the pair process on $[2] \times [2] \cong [4]$ has conditional family the product $V = A \otimes B$ with product laws, and its deficit factorizes through flatness of A:

$$\Delta_V(2,0,1) \;=\; A(2 \leftarrow 0) \otimes \Delta_B(2,0,1), \qquad \|\Delta_V\|^2 = \|A(2 \leftarrow 0)\|^2 \cdot \|\Delta_B\|^2 = \frac{17}{16} \cdot \frac{18}{23575} = \frac{153}{188600},$$

so $\|\Delta_V\| > 0$ and $h_4(V) > 0$ by (7). Now cut by the first-coordinate projection $\lambda(v_A, v_B) = v_A$. The compression returns the A family exactly, $V^\lambda = A$, which is flat: $h_2(V^\lambda) = 0$. Coarse-graining destroyed the charge, and the mechanism is visible in advance: the discarded factor is autonomous and carries all the memory. Proposition 6.4 shows this mechanism is the general one and proves that the flux of the fine deficit through this cut vanishes identically: the deficit is entirely severed.

Example 3.4 (A closed cut, for contrast). Replace M in Example 3.2 by

$$M_{\mathrm{sl}} = \begin{pmatrix} 1/2 & 1/4 & 1/8 \\ 1/4 & 1/2 & 1/8 \\ 1/4 & 1/4 & 3/4 \end{pmatrix},$$

which is strongly lumpable for the same partition (exit mass into $\{3\}$ equals $1/4$ from both states of the block). The coarse family is exactly flat: the same cut that produced charge in Example 3.2 produces none here. Theorem 4.5 explains both instances by one criterion.

4 The balance law

4.1 Flux, production, and the severed component

Definition 4.1. Let (V, p) be a family on $[N]$ with strictly positive laws and λ a cut. For $s < r < t$ define

$$\begin{aligned} \text{the } \textit{flux} \qquad & \Phi_\lambda(t,s,r) \;=\; \mathcal{C}_s\big[\Delta_V(t,s,r)\big] \;=\; L\,\Delta_V(t,s,r)\,R_s, \\ \text{the } \textit{production} \qquad & \mathrm{P}_\lambda(t,s,r) \;=\; L\,V(t \leftarrow r)\,Q_r\,V(r \leftarrow s)\,R_s, \\ \text{the } \textit{severed deficit} \qquad & \Xi_\lambda(t,s,r) \;=\; \Delta_V(t,s,r) \;-\; \iota_s\big[\Phi_\lambda(t,s,r)\big]. \end{aligned}$$

All three have vanishing column sums; flux and production are coarse objects measured in the induced metric; the severed deficit is a fine object. Norms of the cochains are the triple sums as in Section 2.

Theorem 4.2 (Balance law)**.** *For every cut compatible with the law and all $s < r < t$:*

(i) (Balance) $\Delta_{V^\lambda}(t,s,r) \;=\; \Phi_\lambda(t,s,r) \;+\; \mathrm{P}_\lambda(t,s,r)$, *exactly.*

(ii) (Conservation across the cut) $\|\Delta_V\|^2 \;=\; \|\Phi_\lambda\|^2 \;+\; \|\Xi_\lambda\|^2$*: the fine deficit splits orthogonally into its transmitted and severed components, and in particular the flux is contractive,* $\|\Phi_\lambda\| \le \|\Delta_V\|$*, with equality if and only if nothing is severed.*

(iii) (A priori bounds) $\big|\,\|\mathrm{P}_\lambda\| - \|\Delta_{V^\lambda}\|\,\big| \;\le\; \|\Phi_\lambda\| \;\le\; \|\Delta_V\|$*: creation of coarse deficit is bounded by production, destruction by transmission.*

Proof. (i) Insert $I = \Pi_r + Q_r$ at the cut time:

$$\Delta_{V^\lambda}(t,s,r) = LV(t \leftarrow s)R_s - LV(t \leftarrow r)R_r\, LV(r \leftarrow s)R_s = L\big[V(t \leftarrow s) - V(t \leftarrow r)\Pi_r V(r \leftarrow s)\big]R_s,$$

and $V(t \leftarrow s) - V(t \leftarrow r)\Pi_r V(r \leftarrow s) = \Delta_V(t,s,r) + V(t \leftarrow r)Q_r V(r \leftarrow s)$. (ii) is Lemma 2.3(c) applied triple by triple, using the isometry of ι_s to identify $\|\mathcal{C}_s[\Delta_V]\|$ in the induced metric with the norm of the block-measurable component. (iii) is the triangle inequality applied to (i), then (ii). □

Remark 4.3. The theorem is the promised bookkeeping. The fine deficit does not pass to the coarse level intact: part of it, Ξ_λ, lives on directions no coarse observable can see and is lost at the cut; the rest, Φ_λ, is transmitted without amplification. Everything in the coarse deficit beyond the transmitted part is manufactured at the cut itself by the production term, which involves the family only through its passage across the fluctuation subspace at the cut time. Example 3.2 is pure production (the fine deficit vanishes, so flux and severed part vanish and $\Delta_{V^\lambda} = \mathrm{P}_\lambda$, an identity (9) realizes exactly); Example 3.3 is pure severing (the production vanishes by Proposition 6.4, and the flux of the fine deficit is identically zero, so the entire charge is severed). The general cut mixes the two, and Section 7 tabulates the pure regimes realized exactly by the families of this paper.

4.2 The zero set of production: dynamically closed cuts

Definition 4.4. The cut λ is *dynamically closed* for (V,p) at level r if

$$L\,V(t \leftarrow r)\,Q_r \;=\; 0 \qquad \text{for every } t > r,$$

that is, if the conditional law of the coarse future given the fine present, $u \mapsto \big(LV(t \leftarrow r)\big)_{\cdot u}$, is constant on the blocks of λ. The cut is dynamically closed if it is closed at every level $0 < r < m$. The *accessible subspace* at level r is the linear span $\mathcal{A}_r = \mathrm{span}\{\, V(r \leftarrow s)R_s\, e_a :\ s < r,\ a \in [n]\,\} \subseteq \mathbb{R}^N$.

Theorem 4.5 (Zero set of production)**.** *For every cut λ compatible with the law:*

(i) if λ is dynamically closed (Definition 4.4) then $\mathrm{P}_\lambda \equiv 0$;

(ii) conversely, $\mathrm{P}_\lambda \equiv 0$ if and only if $L\,V(t \leftarrow r)\,Q_r$ vanishes on the accessible subspace $\mathcal{A}_r$ for all $0 < r < t \le m$: production vanishes exactly when the cut is dynamically closed on the directions the process reaches.

Proof. (i) Each production term contains the factor $LV(t \leftarrow r)Q_r$ on the left of $V(r \leftarrow s)R_s$. (ii) The a-th column of $\mathrm{P}_\lambda(t,s,r)$ is $LV(t \leftarrow r)Q_r$ applied to the vector $V(r \leftarrow s)R_s\, e_a$, and as $s < r$ and $a \in [n]$ vary these vectors span $\mathcal{A}_r$ by definition; so $\mathrm{P}_\lambda(t,s,r) = 0$ for all $s < r < t$ with cut time r if and only if $LV(t \leftarrow r)Q_r$ vanishes on $\mathcal{A}_r$ for all $t > r$. □

Corollary 4.6 (Weak lumpability recovered)**.** *Let X be a process with strictly positive one-time laws whose conditional family V is flat, as is the case for a Markov chain, and let λ be a cut. The following are equivalent:*

(i) $\mathrm{P}_\lambda \equiv 0$;

(ii) V^λ is flat, i.e., $h_n(V^\lambda) = 0$;

(iii) the two-time conditionals of $Y = \lambda(X)$ satisfy the Chapman–Kolmogorov relation on the grid.

Each is implied by Markovianity of Y (weak lumpability of (X,λ) at the law p_0 [5, 13]), and by strong lumpability of the one-step kernels for the partition, $LM_k = \mathsf{m}_k L$ for every k and some column-stochastic m_k, where $M_k = V(k \leftarrow k{-}1)$, which forces dynamical closure at every level. Thus the zero set of production is weak lumpability in its two-time formulation: the coarse family composes although the coarse process need not be Markov.

Proof. Flatness of V kills the flux, so $\Delta_{V^\lambda} = \mathrm{P}_\lambda$ by Theorem 4.2(i), giving (i)⇔(ii); (ii)⇔(iii) is Proposition 2.2 and the definition of flatness. If Y is Markov its conditional family composes, which is (iii). For strong lumpability [5], $LM_k = \mathsf{m}_k L$ for all k gives $LV(t \leftarrow r) = LM_t \cdots M_{r+1} = \mathsf{m}_t \cdots \mathsf{m}_{r+1} L$, and $LQ_r = L - LR_rL = L - L = 0$, so $LV(t \leftarrow r)Q_r = \mathsf{m}_t \cdots \mathsf{m}_{r+1} LQ_r = 0$: the cut is dynamically closed. Example 3.4 is this case. That (iii) does not imply Markovianity of Y is the standard gap between pairwise Chapman–Kolmogorov and the Markov property; the delicacy of the full weak-lumpability set is classical [13, 12]. □

Corollary 4.7 (Charge-level balance)**.** *With L_*, C_*, θ as in (7) at dimension n,*

$$\tfrac{1}{L_*}\Big(\|\mathrm{P}_\lambda\| - \|\Delta_V\|\Big)_+ \;\le\; h_n(V^\lambda) \;\le\; C_*\big(\|\Phi_\lambda\| + \|\mathrm{P}_\lambda\|\big)^\theta,$$

and for a flat fine family

$$\tfrac{1}{L_*}\,\|\mathrm{P}_\lambda\| \;\le\; h_n(V^\lambda) \;\le\; C_*\,\|\mathrm{P}_\lambda\|^\theta :$$

the charge of the observed family measures the production of the cut, two-sidedly.

Proof. Theorem 4.2(i), the triangle inequality, and (7). □

5 Production as a projection-free Mori–Zwanzig kernel

The production term has the unmistakable shape of a memory kernel: resolved propagation to the cut, a fluctuation insertion, resolved observation after. This section makes the identification exact and explains in what sense it is free of the choice that the projection-operator formalism ordinarily requires.

5.1 The discrete Dyson expansion

Proposition 5.1 (Generalized Langevin structure of the coarse family)**.** *Let V be flat with one-step kernels $M_k = V(k \leftarrow k{-}1)$ and let λ be a cut. For $a < b$ define the* segment kernels

$$G(b \leftarrow a) \;=\; L\,M_b\,Q_{b-1}\,M_{b-1}\,Q_{b-2}\cdots Q_{a+1}\,M_{a+1}\,R_a,$$

so that $G(a{+}1 \leftarrow a) = \mathsf{m}_{a+1} = LM_{a+1}R_a$ is the coarse one-step kernel. Then for all $i < j$,

$$V^\lambda(j \leftarrow i) \;=\; \sum_{i=a_0<a_1<\cdots<a_k=j} G(a_k \leftarrow a_{k-1}) \cdots G(a_1 \leftarrow a_0). \tag{10}$$

The word with all segments of length one is the flat part $\mathsf{m}_j \cdots \mathsf{m}_{i+1}$; every longer segment is a memory kernel carrying at least one fluctuation insertion. On the three-point window $s < r < t$ the deficit is the single-insertion kernel exactly:

$$\Delta_{V^\lambda}(t,s,r) \;=\; L\,V(t \leftarrow r)\,Q_r\,V(r \leftarrow s)\,R_s \;=\; \mathrm{P}_\lambda(t,s,r). \tag{11}$$

Proof. Write $V^\lambda(j \leftarrow i) = LM_j\Big(\prod_{k=j-1}^{i+1}(\Pi_k+Q_k)M_k\Big)R_i$ and expand multilinearly over the choices Π_k versus Q_k. Each occurrence of $\Pi_k = R_kL$ splits its word into two factors joined at time k; grouping the factors between consecutive Π's (and the endpoints) yields exactly the segment kernels, and the sum over binary words is the sum over compositions $i = a_0 < \cdots < a_k = j$ of the interval. Equation (11) is Theorem 4.2(i) with vanishing flux, or directly: insert $I = \Pi_r + Q_r$ once. □

Equation (10) is the discrete generalized Langevin expansion of the projected dynamics, the finite-state analog of the Nakajima–Zwanzig equation [9, 21, 8, 4, 22] in the form used by optimal prediction and by data-driven model reduction [3, 6]; the elimination-of-variables reading is classical [19]. The deficit (11) is its leading, single-insertion shadow at the level of two-time kernels: the *fluctuation current* across the cut, the portion of the $s \to t$ transition current that crosses the cut time inside the subspace invisible to the coarse description. In the language of open quantum systems, where the failure of divisibility of a reduced dynamical map plays the same role [1, 10, 7], production is the classical commutative instance of information backflow sourced at the system–environment cut.

5.2 Independence of the projection operator

The Mori–Zwanzig formalism begins by choosing a projection: Zwanzig's conditional expectation, Mori's linear projection onto a few observables, the Wiener projections of [6]. Different choices give different memory kernels for the same reduction, and the dependence is a standing objection to attaching physical meaning to any one kernel. The present setting dissolves the dependence, in the following precise sense.

Theorem 5.2 (Projection-freeness)**.** *Let V be flat with laws p and λ a cut.*

(i) The deficit Δ_{V^λ} and the charge $h_n(V^\lambda)$ are functionals of the observed family V^λ alone; no projection enters their definition.

(ii) The memory kernel built from the law projection Π_r reproduces the deficit exactly: (11)*.*

(iii) Let $\widetilde{R}_r$ be any admissible section ($L\widetilde{R}_r = I_n$, columns non-negative and supported on the blocks), with idempotent $\widetilde{\Pi}_r = \widetilde{R}_rL$ and kernel $\widetilde{\mathrm{P}}(t,s,r) = LV(t \leftarrow r)(I - \widetilde{\Pi}_r)V(r \leftarrow s)R_s$. Then

$$\widetilde{\mathrm{P}}(t,s,r) \;-\; \Delta_{V^\lambda}(t,s,r) \;=\; L\,V(t \leftarrow r)\,\big(R_r - \widetilde{R}_r\big)\,V^\lambda(r \leftarrow s)\,:$$

the kernel of any other projection differs from the deficit by the mismatch of its resolved model with the observed conditionals, and, with the time-r law, and hence R_r, held fixed, the difference vanishes for every flat family with that time-r law if and only if $\widetilde{R}_r = R_r$. The law projection is thus the unique admissible choice whose resolved model is the observed family, and its kernel is the projection-free object of (i).

Proof. (i) is the definition of the deficit and of (3). (ii) is (11). (iii): subtract $\Delta_{V^\lambda} = LV(t \leftarrow r)(I - R_rL)V(r \leftarrow s)R_s$ from $\widetilde{\mathrm{P}}$ and use $LV(r \leftarrow s)R_s = V^\lambda(r \leftarrow s)$; the difference is $LV(t \leftarrow r)(R_r - \widetilde{R}_r)\,V^\lambda(r \leftarrow s)$. For the uniqueness claim, note first that $L(R_r - \widetilde{R}_r) = 0$ always, so the discrepancy is invisible to families that do not separate states within a block. If $\widetilde{R}_r \neq R_r$, pick a block a on which the sections differ and a state u_0 of that block with $d(u_0) \neq 0$, where $d = (R_r - \widetilde{R}_r)e_a$, a nonzero vector supported on the block with $\sum_u d(u) = 0$. Build a flat family, carrying the given time-r law at every time, from one-step kernels: the identity on every step up to time r, so that $V(r \leftarrow s) = I$ and $V^\lambda(r \leftarrow s) = I_n$, and, on the step leaving r, the kernel sending u_0

into one coarse block b and every other state of $[N]$ into a different block b'; then, at $t = r+1$, the b-entry of the a-th column of the difference is $\big(LV(t \leftarrow r)\, d\big)_b = d(u_0) \neq 0$. The detecting kernels may be taken strictly positive by convex perturbation toward a reset kernel $w\mathbf{1}^\top$ with $w > 0$; the reset kernel annihilates d since $\mathbf{1}^\top d = 0$, so the perturbation preserves the nonvanishing of the difference. □

Remark 5.3 (The residual at the projection onto flat families). There is a second, fully variational avatar. Let $W^* \in \mathcal{F}\mathrm{lat}_n$ attain $h_n(V^\lambda)$. The residual $V^\lambda - W^*$ is normal to the flat locus at W^* in the sense of the first-order theory of [18, Section 6] and the normal cones of variational analysis [11]; it is defined with no projection whatsoever, and by (11) together with (7) its size is two-sidedly comparable to that of the Mori–Zwanzig kernel:

$$\tfrac{1}{L_*}\|\mathrm{P}_\lambda\| \;\le\; \|V^\lambda - W^*\| \;=\; h_n(V^\lambda) \;\le\; C_*\|\mathrm{P}_\lambda\|^\theta.$$

The production term is thus the fluctuation current across the cut, its size read, two-sidedly, at the metric projection onto flat families; this is the form the memory kernel takes when the reduced description is required to carry its own laws rather than a chosen projection.

6 Towers and monotonicity

6.1 Towers and the telescoped balance law

A *tower* is a sequence of cuts

$$[N] = [n_0] \xrightarrow{\lambda_1} [n_1] \xrightarrow{\lambda_2} \cdots \xrightarrow{\lambda_K} [n_K], \qquad n_0 > n_1 > \cdots > n_K \geq 2,$$

with families $V^{(0)} = V$ and $V^{(k)} = \big(V^{(k-1)}\big)^{\lambda_k}$, laws lumped accordingly. Because two-time conditionals lump through any composition of cuts (Proposition 2.2), $V^{(k)}$ is also the direct compression of V through $\lambda_k \circ \cdots \circ \lambda_1$: disintegrations compose, $R_i^{\mu\circ\lambda} = R_i^\lambda R_i^\mu$ by direct computation on the nested blocks, so $L_\mu(L_\lambda V R_i^\lambda)R_i^\mu = L_{\mu\circ\lambda} V R_i^{\mu\circ\lambda}$, and the tower and the composed cut give the same family.

Corollary 6.1 (Telescoped balance). *Writing $\mathcal{C}^{(k)}$, $\mathrm{P}^{(k)}$, $\Phi^{(k)}$, $\Xi^{(k)}$ for the objects of Definitions 2.1 and 4.1 at step k,*

$$\Delta_{V^{(K)}} \;=\; \mathcal{C}^{(K)}\cdots\mathcal{C}^{(1)}\big[\Delta_V\big] \;+\; \sum_{k=1}^{K} \mathcal{C}^{(K)}\cdots\mathcal{C}^{(k+1)}\big[\mathrm{P}^{(k)}\big],$$

each compression contractive in its induced metric: the deficit of the fully coarse family is the multiply-compressed fine deficit plus the compressed productions of every intermediate cut.

Proof. Induction on K with Theorem 4.2(i). □

6.2 Monotonicity with sharp hypotheses

Theorem 6.2 (Monotonicity). *Consider one tower step λ applied to (V, p), and suppose*

(S) *nothing is severed: $\Xi_\lambda = 0$, equivalently $\|\Phi_\lambda\| = \|\Delta_V\|$; and*

(C) *production is coherent: $\langle \Phi_\lambda, \mathrm{P}_\lambda \rangle \geq 0$ as cochains in the induced metric.*

Then

$$\|\Delta_{V^\lambda}\|^2 \;=\; \|\Delta_V\|^2 \;+\; 2\langle \Phi_\lambda, \mathrm{P}_\lambda\rangle \;+\; \|\mathrm{P}_\lambda\|^2 \;\geq\; \|\Delta_V\|^2 + \|\mathrm{P}_\lambda\|^2,$$

so the deficit norm is nondecreasing, strictly increasing whenever $\mathrm{P}_\lambda \neq 0$*. Along a tower every step of which satisfies* (S) *and* (C)*, the deficit norm is nondecreasing in* k*; by* (7)*, with the constants of the corresponding dimensions and laws, the charge then satisfies* $h_{n_{k+1}}(V^{(k+1)}) \geq \frac{1}{L_*}\|\Delta_{V^{(k)}}\| \geq \frac{1}{L_* C_*^{1/\theta}}\, h_{n_k}(V^{(k)})^{1/\theta}$*, and the qualitative statements transfer exactly: positivity of the charge is preserved, and the deficit norm can grow only through production. Both hypotheses are sharp, exactly: Example 3.3 violates* (S) *while* (C) *holds trivially, and destroys the charge; Proposition 6.5 violates* (C) *while* (S) *holds exactly, and destroys the charge.*

Proof. By Theorem 4.2(i), $\|\Delta_{V^\lambda}\|^2 = \|\Phi_\lambda + \mathrm{P}_\lambda\|^2 = \|\Phi_\lambda\|^2 + 2\langle \Phi_\lambda, \mathrm{P}_\lambda\rangle + \|\mathrm{P}_\lambda\|^2$; under (S) $\|\Phi_\lambda\|^2 = \|\Delta_V\|^2$ by Theorem 4.2(ii), and (C) removes the cross term's sign ambiguity. The tower statement iterates, and the charge transfer is (7) applied twice. □

Remark 6.3. The theorem localizes non-monotonicity in exactly two mechanisms. The charge can fall only if the cut severs deficit-bearing directions (failure of (S)) or if production interferes destructively with the transmitted deficit (failure of (C)). The first mechanism is geometric and is the general form of "discarding an autonomous factor" (Proposition 6.4); the second is dynamical interference at the cut, and Proposition 6.5 shows it can act alone and act totally. What cannot happen is destruction without one of the two: that is the content of the identity in the theorem, with no hypotheses.

6.3 Autonomous factors and total severing

Proposition 6.4 (Discarding an autonomous factor)**.** *Let the fine process be an independent pair,* $X = (X', X'')$ *on* $[n'] \times [n'']$*, so that* $V = V' \otimes V''$ *with product laws* $p_i = p_i' \otimes p_i''$*, and let* $\lambda(v', v'') = v'$ *be the projection discarding the autonomous factor* X''*. Then:*

(i) the cut is dynamically closed: $\mathrm{P}_\lambda \equiv 0$*;*

(ii) the flux is the retained factor's own deficit: $\Phi_\lambda(t, s, r) = \Delta_{V'}(t, s, r)$*, and* $V^\lambda = V'$*;*

(iii) consequently $\|\Xi_\lambda\|^2 = \|\Delta_V\|^2 - \|\Delta_{V'}\|^2$*: the entire memory of the discarded factor, including its cross terms with the retained one, is severed. If the retained factor is flat and the discarded factor carries deficit, the flux vanishes identically and the whole charge is destroyed, which is Example 3.3.*

Proof. With product laws, $\Pi_r = I_{n'} \otimes p_r'' \mathbf{1}^\top$ and $Q_r = I_{n'} \otimes (I_{n''} - p_r'' \mathbf{1}^\top)$. (i) $LV(t \leftarrow r)Q_r = V'(t \leftarrow r) \otimes \big[\mathbf{1}^\top V''(t \leftarrow r)(I - p_r'' \mathbf{1}^\top)\big]$, and $\mathbf{1}^\top V''(t \leftarrow r) = \mathbf{1}^\top$ gives $\mathbf{1}^\top (I - p_r'' \mathbf{1}^\top) = \mathbf{1}^\top - \mathbf{1}^\top = 0$. (ii) For any kernels A on $[n']$ and B on $[n'']$, $\mathcal{C}_s[A \otimes B] = A \cdot \big(\sum_u p_s''(u) \sum_v B_{vu}\big)$; the deficit of a product with laws splits as $\Delta_V = \Delta_{V'} \otimes V''(t \leftarrow s) + \big(V'(t \leftarrow r)V'(r \leftarrow s)\big) \otimes \Delta_{V''}$; the first factor's companion $V''(t \leftarrow s)$ has unit column sums and the second's companion $\Delta_{V''}$ has zero column sums, so the compression returns $\Delta_{V'} \cdot 1 + \big(V'V'\big) \cdot 0$. (iii) is Theorem 4.2(ii) with (ii). □

The proposition makes precise what it means for a tower to discard an autonomous factor: some step projects away an independent subdynamics, and then hypothesis (S) fails by exactly the discarded factor's share of the deficit. The condition is sharp in the strong sense that the destruction counterexample realizes it with all quantities exact.

6.4 Perfect interference: sharpness of the coherence hypothesis

Hypothesis (S) does not suffice on its own. The following construction produces a process whose cut severs nothing and discards nothing, and whose production term is *exactly opposite* to the transmitted deficit, so that the two annihilate and the coarse family is flat.

Proposition 6.5 (Perfect interference)**.** *There is a process-induced family on three states over the grid* $\{0,1,2\}$*, with an explicitly displayed strictly positive joint law, and a cut, such that*

(i) nothing is severed: $\Xi_\lambda = 0$*, and no autonomous factor is discarded, since the cut is not dynamically closed;*

(ii) production is nonzero and maximally incoherent: $\mathrm{P}_\lambda = -\Phi_\lambda$*, so* $\langle \Phi_\lambda, \mathrm{P}_\lambda \rangle = -\|\Phi_\lambda\|\,\|\mathrm{P}_\lambda\| < 0$*, equality in Cauchy–Schwarz with the wrong sign;*

(iii) consequently $\Delta_{V^\lambda} = \Phi_\lambda + \mathrm{P}_\lambda = 0$*: the coarse family is flat, and the positive charge of the fine family is destroyed entirely by interference.*

Hypothesis (C) *therefore cannot be dropped from Theorem 6.2 even when* (S) *holds exactly.*

Proof. All laws are uniform, $p_0 = p_1 = p_2 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})^\top$, and the cut is $\lambda(1) = \lambda(2) = a$, $\lambda(3) = b$, with coarse laws $(\frac{2}{3}, \frac{1}{3})$. Take the doubly stochastic one-step kernels

$$A = V(1 \leftarrow 0) = \begin{pmatrix} 1/2 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \end{pmatrix}, \qquad B = V(2 \leftarrow 1) = \begin{pmatrix} 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \\ 1/2 & 1/4 & 1/4 \end{pmatrix},$$

and perturb the long kernel off the Markov product by a block lift: with

$$\delta_0 = \begin{pmatrix} 1 & -2 \\ -1 & 2 \end{pmatrix}, \qquad \iota_0[\delta_0] = \begin{pmatrix} 1/2 & 1/2 & -1 \\ 1/2 & 1/2 & -1 \\ -1 & -1 & 2 \end{pmatrix},$$

$$V(2 \leftarrow 0) \;=\; BA + \tfrac{1}{64}\,\iota_0[\delta_0] = \begin{pmatrix} 41/128 & 49/128 & 19/64 \\ 41/128 & 41/128 & 23/64 \\ 23/64 & 19/64 & 11/32 \end{pmatrix},$$

which is entrywise positive and column-stochastic, and preserves the uniform law, since $\iota_0[\delta_0]$ has vanishing column and row sums. The line $\mathbb{R}\,\delta_0$ is exactly the set of coarse kernel differences with vanishing column sums whose block lifts preserve the uniform law for this partition, so the construction stays inside marginal-consistent families.

The family is process-induced. Define the joint law on $[3]^3$ by

$$\pi(u,v,w) \;=\; \tfrac{1}{3}\,A_{vu}B_{wv} \;+\; \tfrac{1}{9}\,\iota_0[\delta_0]_{wu} \cdot \tfrac{1}{64} \;\geq\; \tfrac{1}{48} - \tfrac{1}{576} \;=\; \tfrac{11}{576} \;>\; 0.$$

Summing over w kills the correction column-wise and returns the $(0,1)$ margin $\frac{1}{3}A_{vu}$; summing over u kills it row-wise and returns the $(1,2)$ margin $\frac{1}{3}B_{wv}$; summing over v returns $\frac{1}{3}\big(BA + \frac{1}{64}\iota_0[\delta_0]\big)_{wu}$, the $(0,2)$ margin. The three two-time conditional kernels of the process with law π are therefore exactly A, B, and $V(2 \leftarrow 0)$, with uniform one-time laws.

Nothing is severed. The single fine deficit is $\Delta_V(2,0,1) = V(2 \leftarrow 0) - BA = \frac{1}{64}\iota_0[\delta_0]$, a block lift; hence $\Phi_\lambda = \mathcal{C}_0[\Delta_V] = \frac{1}{64}\delta_0$ and $\Xi_\lambda = \Delta_V - \iota_0[\Phi_\lambda] = 0$ by Lemma 2.3(b). The fine charge is positive: $\|\Delta_V\|^2 = \|\frac{1}{64}\delta_0\|^2 = \frac{9}{4096}$, so $\|\Delta_V\| = \frac{3}{64}$ and $h_3(V) > 0$.

Production. With $\Pi_1 = R_1 L$ for the uniform law, $Q_1 = I - \Pi_1$ averages the first two coordinates against each other: $Q_1 A$ has rows $\frac{1}{2}(\mathrm{row}_1 - \mathrm{row}_2)(A) = (1/8,\, 0,\, -1/8)$, its negative, and zero. Then

$$B\,Q_1 A = \begin{pmatrix} 0 & 0 & 0 \\ -1/32 & 0 & 1/32 \\ 1/32 & 0 & -1/32 \end{pmatrix}, \qquad \mathrm{P}_\lambda(2,0,1) = L\,B\,Q_1 A\,R_0 = \begin{pmatrix} -1/64 & 1/32 \\ 1/64 & -1/32 \end{pmatrix} = -\tfrac{1}{64}\,\delta_0,$$

which is nonzero, so the cut is not dynamically closed and, by Proposition 6.4(i), the cut discards no autonomous factor. Thus $\mathrm{P}_\lambda = -\Phi_\lambda$, the balance law gives $\Delta_{V^\lambda} = \Phi_\lambda + \mathrm{P}_\lambda = 0$, and one checks directly that $LV(2 \leftarrow 0)R_0 = \big(LBR_1\big)\big(LAR_0\big)$. The cross term is $\langle \Phi_\lambda, \mathrm{P}_\lambda \rangle = -\|\frac{1}{64}\delta_0\|^2 = -\frac{9}{4096}$, maximally incoherent. Every equality above is an identity of rational matrices. □

Remark 6.6 (The interference dial). The construction is a one-parameter family: replace $\frac{1}{64}$ by x in $V(2 \leftarrow 0) = BA + x\,\iota_0[\delta_0]$ (admissible for an interval of x around the origin, with the joint law positive for $|x|$ small). Production depends only on A, B, and the cut, so $\mathrm{P}_\lambda = -\frac{1}{64}\delta_0$ for every x, while $\Phi_\lambda = x\,\delta_0$; the balance law gives, in closed form,

$$\|\Delta_V\| \;=\; 3\,|x|, \qquad \|\Delta_{V^\lambda}\| \;=\; 3\,\big|x - \tfrac{1}{64}\big|.$$

The cut subtracts a fixed quantum of deficit from whatever is transmitted. For $x < 0$ production is coherent and the deficit grows, at $x = 0$ the fine family is flat and the cut is pure production, at $x = \frac{1}{128}$ the deficit norm is exactly conserved, at $x = \frac{1}{64}$ it is annihilated, and beyond that the coarse deficit grows again with reversed orientation. One family walks the entire phase diagram of the balance law.

Remark 6.7. Proposition 6.5 also shows that for charged fine families, flatness of the observed family does not certify a closed cut: here V^λ is exactly flat while $\mathrm{P}_\lambda \neq 0$. The equivalence of Corollary 4.6 between vanishing production and coarse flatness is a feature of flat fine families and does not survive upstream charge; what survives in general is only the balance law itself.

7 Computations

Every quantity in this paper is computed in exact rational arithmetic; there is no numerical optimization, no simulation, no data, and no floating point anywhere in the argument. Positivity of charges is certified by (7) together with exact positivity of deficit norms, and flatness by exact vanishing, since $\mathcal{F}\mathrm{lat}_n$ is precisely the zero set of the deficit. The metric is (6) with the families' own laws and unit pair and triple weights. Table 1 gathers the four families worked in the text; each realizes one pure regime of the balance law, and each row satisfies the balance identity $\|\Delta_{V^\lambda}\|^2 = \|\Phi_\lambda\|^2 + 2\langle \Phi_\lambda, \mathrm{P}_\lambda \rangle + \|\mathrm{P}_\lambda\|^2$ and the conservation identity $\|\Delta_V\|^2 = \|\Phi_\lambda\|^2 + \|\Xi_\lambda\|^2$ as identities of rational numbers, which the reader can confirm by inspection of the rows.

The table is the phase diagram of Theorem 4.2 read at its corners. The first row creates deficit from a chargeless family through production alone; the third row is the same cut geometry made dynamically closed, and everything vanishes. The second row destroys the charge with zero production, through pure severing, and its fine entry factorizes exactly as the χ^2-size of the flat companion kernel times the discarded factor's deficit, $153/188600 = (17/16) \cdot (18/23575)$, so the row is arithmetically tied to the first. The fourth row is the only one with a nonzero cross term, and the cross term is extremal, $\langle \Phi_\lambda, \mathrm{P}_\lambda \rangle = -\|\Phi_\lambda\|\|\mathrm{P}_\lambda\|$: transmission and production annihilate. The dial of Remark 6.6 moves a single family through the first and fourth rows and everything in between, with the closed forms $\|\Delta_V\| = 3|x|$ and $\|\Delta_{V^\lambda}\| = 3\,|x - \frac{1}{64}|$.

Table 1: The four pure regimes of the balance law. All entries are exact; squared norms and inner products are taken in the metric (6) at the appropriate level, deficit cochains summed over triples.

Family	$\|\Delta_V\|^2$	$\|\Phi_\lambda\|^2$	$\|\Xi_\lambda\|^2$	$\|\mathrm{P}_\lambda\|^2$	$\langle\Phi_\lambda, \mathrm{P}_\lambda\rangle$	$\|\Delta_{V^\lambda}\|^2$
Ex. 3.2 (production)	0	0	0	18/23575	0	18/23575
Ex. 3.3 (severing)	153/188600	0	153/188600	0	0	0
Ex. 3.4 (closed cut)	0	0	0	0	0	0
Prop. 6.5 (interference)	9/4096	9/4096	0	9/4096	−9/4096	0

8 Open problems

(i) *Coherent cuts.* Characterize the cuts for which $\langle\Phi_\lambda, \mathrm{P}_\lambda\rangle \geq 0$ structurally, rather than instance by instance: is there a geometric relation between the orientation of the incoming deficit and the fluctuation subspace of the cut that decides the sign? Proposition 6.5 shows incoherence can be total, and Remark 6.6 moves a single family continuously from coherent to maximally incoherent, so the answer is certainly not "always coherent"; what is missing is the dividing geometry.

(ii) *Balance for the full profile.* This paper works at the dimension-matched level h_n. For $d > n$ the sets $\mathcal{F}_d$ of [18] are strictly larger than $\mathcal{F}\mathrm{lat}_n$, and coarse-graining should obey a balance law for h_d in which production is sourced only by the part of the cut not explained by $d-n$ hidden states; the memory rank of the lumped family against the rank of the fine one is the discrete shadow. Formulate and prove it.

(iii) *Continuous grids and production rates.* On refining grids the production over a window should scale to a production *rate*, and the balance law to a continuity equation for memory along the coarse-graining semigroup, with dynamically closed cuts as the vanishing-rate ideal. The formal analogy with entropy production along a cut in stochastic thermodynamics deserves to be made exact or refuted.

(iv) *Estimation and a lumpability test.* Flux and production are Lipschitz in the family by construction, so plug-in estimates inherit the convergence of the family estimator as in [18, Proposition 4.6]. A test of $\mathrm{P}_\lambda = 0$ against the two-sided window of Corollary 4.7 is a practical test of dynamical closure, hence of two-time weak lumpability, from data; its null distribution is unexplored.

(v) *Optimal cuts.* Among cuts of fixed coarse dimension, minimizing production (equivalently, by Corollary 4.7, the charge of the observed family when the fine family is flat) is a model-reduction objective dual to the optimal-prediction program [3, 6]: it seeks the most dynamically autonomous coarse description rather than the most predictive one. Quantifying the gap between the two optima, and the relation of production-minimizing cuts to the spectral structure of the fine generator when one exists, is open.

Declarations

AI assistance was used in the preparation of this manuscript. All theoretical frameworks, methodological approaches, and conclusions are the author's original work; the author reviewed all content and assumes responsibility for it.

Competing interests

The author declares no competing interests. Funding information appears in the footnote to the title.

Data and code availability

This paper contains no numerical results, no simulations, and no external data. Every computation is exact rational arithmetic on the matrices displayed in the text, and every claimed identity or inequality can be verified by hand from those matrices.

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